

Computing a chemical potential with a biasing potential in the reference system

manuel.athenes

November 2024

1 Introduction

Definitions: let $p = (\mathbf{p}_1, \dots, \mathbf{p}_{N+1})^T$ and $r = (\mathbf{r}_1, \dots, \mathbf{r}_{N+1})^T$ denote the momenta and positions of $N + 1$ particles. We note $U^N(r)$ and $U^{N+1}(r)$ the potential energy for which N and $N + 1$ particles are physically interacting, respectively. In potential energy $U^N(r)$, particle $\mathbf{r}_{N+1}$ is a ghost non-interacting particle. We denote the two systems by S_0 and S_1 , respectively, and their Hamiltonians by $H^N(r, p) = U^N(r, p) + K$ and $H^{N+1} = U^{N+1}(r) + K(p)$, where $K(p) = p^T M^{-1} p$ is the kinetic energy of the $(N + 1)$ st particle.

Free energy of ideal gas: An ideal gas is a system of non-interacting particles. The ideal gas contribution to the free energy of $N + 1$ particles is

$$F_{\text{id}} = -k_B T \ln \left(\frac{1}{h^{N+1}} \prod_{\alpha} \frac{1}{N_{\alpha}!} \int \exp[-\beta K(p)] dr dp \right) \quad (1)$$

$$= -k_B T \ln \left(\sqrt{\det \left(\frac{2\pi k_B T}{h^2} M \right)} \prod_{\alpha} \frac{V^{N_{\alpha}}}{N_{\alpha}!} \right) \quad (2)$$

where V is the volume of the computational cell, T is temperature, $\beta = 1/k_B T$ and N_{α} the number of particles of type α . Note that $\sum_{\alpha} N_{\alpha} = N + 1$. The mass matrix being diagonal with diagonal components equal to m_{ν} , we have

$$\sqrt{\det \left(\frac{2\pi k_B T}{h^2} M \right)} = \prod_{\nu=1}^{3N+3} \sqrt{\frac{2\pi k_B T}{h^2} m_{\nu}} = \prod_{\alpha} \Lambda_{\alpha}^{-3N_{\alpha}} \quad (3)$$

where Λ_{α} is de Broglie's thermal wavelength of particle of type α .

Free energy: The free energy of particle system S_X ($X = 0, 1$) is defined by

$$F^{N+X} = -k_B T \ln \left(\frac{1}{h^{N+1}} \prod_{\alpha} \frac{1}{N_{\alpha}!} \int \exp[-\beta H^{N+X}(r, p)] dr dp \right) \quad (4)$$

$$= F_{\text{id}} - k_B T \ln \left(\frac{1}{V^{N+1}} \int \exp[-\beta U^{N+X}(r)] dr \right) \quad (5)$$

Excess free energy and chemical potential: The excess free energy of system S_X ($X = 0, 1$) is the contribution arising from physical interactions between the particles, i.e. the free energy minus the ideal contribution

$$F_{\text{ex}}^{N+X} = F^{N+X} - F_{\text{id}} = -k_B T \ln \left(\frac{1}{V^{N+1}} \int \exp[-\beta U^{N+X}(r)] dr \right). \quad (6)$$

The excess free energy is zero for an ideal gas. The excess chemical potential of particle $N + 1$ is

$$\mu_{\text{ex}} = F_{\text{ex}}^{N+1} - F_{\text{ex}}^N \quad (7)$$

and can be expressed using the logarithmic ratio of two partition functions

$$\mu_{\text{ex}} = -k_B T \ln \frac{\int \exp[-\beta U^{N+1}(r)] dr}{\int \exp[-\beta U^N(r)] dr} \quad (8)$$

Auxiliary biasing potential and correction to free energy differences We consider that the $N + 1$ -st particle undergoes the action of an auxiliary biasing potential whose aim is to confine the particle to a specific region of the position space centered around $\mathbf{r}_c$. The biasing potential is chosen to be harmonic

$$U_{\text{har}}(r) = \frac{\kappa}{2} \|\mathbf{r}_{N+1} - \mathbf{r}_c\|^2 \quad (9)$$

and is added to the reference system, now denoted by $\tilde{S}_0$. The Hamiltonian of the reference system now becomes

$$\tilde{H}^N(r, p) = H^N(r, p) + U_{\text{har}}(r) \quad (10)$$

$$\tilde{H}^N(r, p) = H^N(r, p) + U_{\text{bias}}(r) \quad (11)$$

Since the potential energy of particle $N + 1$ is independent of the N first particles, its contribution to the excess free energy can be integrated separately

$$\tilde{F}_{\text{ex}}^N = -k_B T \ln \left(\frac{1}{V^N} \int \exp[-\beta U^N(\tilde{r})] d\mathbf{r}_1 d\mathbf{r}_2 \cdots d\mathbf{r}_N \right) \quad (12)$$

$$-k_B T \ln \left(\frac{1}{V} \int \exp[-\beta U_{\text{har}}(r - \tilde{r})] d\mathbf{r}_{N+1} \right) \quad (13)$$

where $\tilde{r} = (\mathbf{r}_1, \dots, \mathbf{r}_N, \mathbf{0})^T$. The effect of the biasing potential on the excess free energy can easily be evaluated via analytical integration resorting to the error function. Assuming that the position $\mathbf{r}_c$ corresponding to the minimum of the harmonic confining potential is at the center of the cubic simulation cell of dimension $L = \sqrt[3]{V}$, we have

$$\tilde{F}_{\text{ex}}^N = F_{\text{ex}}^N - 3k_{\text{B}}T \ln \left(\frac{1}{L} \sqrt{\frac{2\pi k_{\text{B}}T}{\kappa}} \text{erf} \left(\frac{L}{2} \sqrt{\frac{\kappa}{2k_{\text{B}}T}} \right) \right) \quad (14)$$

$$\tilde{F}_{\text{ex}}^N = F_{\text{ex}}^N + F_{\text{bias}}^1 \quad (15)$$

where we resorted to the integral $\int_{-L/2}^{L/2} \exp[-cx^2] dx = \sqrt{\frac{\pi}{c}} \text{erf}(\sqrt{c}L/2)$ with $c = \kappa/2k_{\text{B}}T$. The excess chemical potential is now given by the following relationship

$$\mu_{\text{ex}} = F_{\text{ex}}^{N+1} - \tilde{F}_{\text{ex}}^N - 3k_{\text{B}}T \ln \left(\frac{1}{L} \sqrt{\frac{2\pi k_{\text{B}}T}{\kappa}} \text{erf} \left(\frac{L}{2} \sqrt{\frac{\kappa}{2k_{\text{B}}T}} \right) \right) \quad (16)$$

In practice this correction is not valid when exclusion volumes exist around atoms. In this case it is more difficult to estimate. The best way is then to simply calculate for a given spring the free energy differences between various sites of the spring and thus discard this correction altogether.

Limiting cases We can check that

$$\lim_{\kappa \rightarrow 0} \tilde{F}_{\text{ex}}^N = F_{\text{ex}}^N \quad (17)$$

because $\text{erf}(x/2) \sim x/\sqrt{\pi}$ in the vicinity of 0. This entails that in absence of harmonic spring attached to particle $N + 1$ (κ tends to $= 0$), the biasing potential is uniformly zero over the cell and the equality $\tilde{F}_{\text{ex}}^N = F_{\text{ex}}^N$ holds. At variance, when the harmonic potential is stiff enough compared to the cell dimension, i.e. $\kappa > 3/L^2$, then the associated normal distribution is almost entirely peaked in the cell and we have

$$\tilde{F}_{\text{ex}}^N \approx F_{\text{ex}}^N + k_{\text{B}}T \ln V - \frac{3k_{\text{B}}T}{2} \ln \frac{2\pi k_{\text{B}}T}{\kappa} \quad (18)$$

because $\ln \text{erf}(x) \sim \exp[-x^2]/2$ in the vicinity of ∞ .

For a stiff enough harmonic biasing potential, the chemical potential of particle $N + 1$ becomes

$$\mu_{\text{ex}} = F_{\text{ex}}^{N+1} - \tilde{F}_{\text{ex}}^N + k_{\text{B}}T \ln V - \frac{3k_{\text{B}}T}{2} \ln \frac{2\pi k_{\text{B}}T}{\kappa} \quad (19)$$

$$\mu_{\text{ex}} = F_{\text{ex}}^{N+1} - \tilde{F}_{\text{ex}}^N + F_{\text{bias}}^1 \quad (20)$$

Extended potential energy and Hamiltonian We introduce the extended potential energy $U_{\lambda}(r, p)$ for which the coupling parameter λ allows to gradually switch from

the reference system to the target one. The extended potential energy and Hamiltonian write

$$U_\lambda(r) = \lambda U^{N+1}(r) + (1 - \lambda)(U_{\text{har}}(r) + U^N(r)) \quad (21)$$

$$H_\lambda(r, p) = p^T M^{-1} p + U_\lambda(r) \quad (22)$$

The additional force that applies to the N+1 atom is

$$\vec{F}_N + 1 = -\kappa(\vec{r} - \vec{r}_c) \quad (23)$$

2 other potentials

slice; grain boundary We consider that the $N + 1$ -st particle undergoes the action of an auxiliary biasing potential whose aim is to confine the particle to a slice of space centered around $\mathbf{x} = \mathbf{x}_c$. The biasing potential is chosen to be harmonic along $\mathbf{x}$ (with L_x the associated cell size)

$$U_{\text{har}}(r) = \frac{\kappa}{2} \|\mathbf{x}_{N+1} - \mathbf{x}_c\|^2 \quad (24)$$

$$F_{\text{bias}}^1 = +k_B T \ln L_x - \frac{k_B T}{2} \ln \frac{2\pi k_B T}{\kappa} \quad (25)$$

The additional force that applies to the N+1 atom is

$$\vec{F}_N + 1 = -\kappa(x - x_c)\vec{X}/X \quad (26)$$

line We consider that the $N + 1$ -st particle undergoes the action of an auxiliary biasing potential whose aim is to confine the particle to a line of space at $y=z=0$. The biasing potential is chosen to be harmonic along the distance to the line

$$U_{\text{har}}(r) = \frac{\kappa}{2} (\|\mathbf{z}_{N+1} - \mathbf{z}_c\|^2 + \|\mathbf{y}_{N+1} - \mathbf{y}_c\|^2) \quad (27)$$

$$F_{\text{bias}}^1 = +k_B T \ln L_y + k_B T \ln L_z - k_B T \ln \frac{2\pi k_B T}{\kappa} \quad (28)$$

bubble We want the insertion to take place inside a bubble (centered in 0). That is we want the probability of insertion to follow this law (fermi-Dirac on the distance to the center with r_b the radius of the bubble and α some smoothing parameter:

$$P_{\text{ins}}(r) = 1 + \exp(\alpha(r - r_b)) \quad (29)$$

Note that P_{ins} is equal to one in favorable area (the bubble) and decreases to 0 outside the bubble. This is important to calculate properly the ratio of volumes in equation 34. It does not contain the r^2 term and it is not normalized. The r^2 term has to be added in the code and the probability must be normalized to build the actual probability of inserting the particle at a certain r .

The potential which induces this probability distribution satisfies

$$P_{\text{ins}}(r) = \exp(-\beta U_{\text{ins}}(r)) \quad (30)$$

$$U_{ins}(r) = -k_B T \ln(P_{ins}(r)) \quad (31)$$

The normalization of P_{ins} is not important here as it only shifts the value of $U_{ins}(r) = -k_B T \ln(1 + \exp(\alpha(r - r_b)))$ (32)

$$F_{bias}^1 = -k_B T \ln\left(\frac{1}{V} \int \exp[-\beta U_{ins}(r - \tilde{r})] d\mathbf{r}\right) \quad (33)$$

$$F_{bias}^1 = -k_B T \ln\left(\frac{1}{V} \int P_{ins}(r) d\mathbf{r}\right) \quad (34)$$

$$F_{bias}^1 = -k_B T \ln\left(\frac{1}{V} \int 4 * \pi * r^2 P_{ins}(r) dr\right) \quad (35)$$

The integral can be calculated numerically. The additional force that applies to the N+1 atom is

$$\vec{F}_N + 1 = -\nabla U_{ins}(r) \quad (36)$$

It can also be evaluated numerically by differences